

The map as one conditional theorem: two hypotheses, both finite in kind, and an explicit lower bound $m \geq \hbar c \gamma' / a_*$

Everything this programme has established about the $SU(3)$ mass gap can be stated as a single conditional theorem. Its two hypotheses are the two things not proved here: control of the blocking steps from the weak side down to the coupling where the correlation length is one lattice unit, and a certified mixing statement at that one coupling on one finite box. Its conclusion is the Jaffe–Witten gap along the trajectory with an explicit lower bound,

$$m \geq \frac{\hbar c \gamma'}{a_*},$$

where a_* is the physical lattice spacing at which the trajectory reaches the verified coupling, about 0.1 to 0.17 fm by the published scale, and γ' is the certified decay rate per lattice unit there. Any positive γ' gives a positive gap, and the data say the true rate is about 0.8. Constants explicit; nothing promoted.

1. Setting

Wilson measure μ_{β_0} on the links of $(a_0\mathbb{Z})^4$ at bare coupling $\beta_0 = 6/g_0^2$, in a periodic box of any size. A block-spin map $\mathcal{B}$ of block size 2: a gauge-covariant assignment of a coarse link variable to each block, measurable in the fine links, such that coarse Wilson loops are functions of fine Wilson loops. The renormalized measures $\mu^{(k)} = \mathcal{B}^k \mu_{\beta_0}$ live on the links of $(2^k a_0 \mathbb{Z})^4$ and, by construction, reproduce exactly the correlations of coarse observables. Write $\Phi^{(k)}$ for the renormalized interaction when it exists, and $\|\Phi\|_\kappa = \sup_\ell \sum_{X \ni \ell} e^{\kappa \text{diam } X} \|\Phi_X\|_\infty$ for the interaction norm with range weight $\kappa > 0$.

2. The hypotheses

H1 (the weak side). There is $K = K(a_0)$ with $2^K a_0 \rightarrow a_*$ as $a_0 \rightarrow 0$, a rate $\gamma_* > 0$ and constants C_k such that for each $k < K$ and all local observables F, G of the k -th lattice with supports at distance d in its units,

$$\left| \langle F; G \rangle_{\mu^{(k)}} - \langle \bar{F}; \bar{G} \rangle_{\mu^{(k+1)}} \right| \leq C_k \|F\|_\infty \|G\|_\infty e^{-\gamma_* d},$$

where $\bar{F}$ is the conditional expectation of F given the coarse links; and $\Phi^{(K)}$ exists, is finite in $\|\cdot\|_\kappa$, and satisfies $\|\Phi^{(K)} - \Phi_0\|_\kappa \leq r$ for a reference interaction Φ_0 and radius r fixed in H2.

In words: at every step before the last, the fluctuations integrated out cluster exponentially at the scale of the step, and the trajectory arrives within a fixed distance of a reference interaction. For the steps at which the effective coupling is small this is the content of the constructive programme, perturbative in kind and unavailable with constants; for the last three, in which ξ/a passes from about 10 to about 1, it is the open problem (bands note §2b).

H2 (one box). The reference interaction Φ_0 satisfies the strong-mixing condition on a box V of side R : for every $W \subseteq V$, every boundary link y and boundary conditions ω, ω' differing at y ,

$$\|\mu_V^\omega|_W - \mu_V^{\omega'}|_W\|_{\text{TV}} \leq C |W| e^{-\gamma d(W,y)},$$

with a margin such that every Φ with $\|\Phi - \Phi_0\|_\kappa \leq r$ satisfies the same condition with some $C', \gamma' > 0$. The natural Φ_0 is the Wilson interaction at $\beta_W \simeq 6$, where the published correlation length is about one lattice unit, and R is 3 to 5; the condition is to be read on the gauge-invariant content of W , the small Wilson loops, since single-link marginals of interior links are Haar (bands note).

3. The theorem

Theorem (conditional). Assume H1 and H2. Then for every bare coupling on the trajectory and every periodic volume:

1. $\mu^{(K)}$ has a unique Gibbs state and truncated correlations of all local observables decaying at rate at least γ' per unit of $2^K a_0$ (Dobrushin–Shlosman; Martinelli–Olivieri);
2. μ_{β_0} has truncated correlations of all local fine observables decaying at rate at least $\min(\gamma_*, \gamma')/2^K$ per unit of a_0 ;
3. the Wilson transfer matrix at spacing a_0 has a spectral gap

$$\Delta(a_0) \geq \frac{\hbar c}{a_0} \cdot \frac{\min(\gamma_*, \gamma')}{2^{K(a_0)}}$$

on the cyclic subspace of local observables, uniformly in the volume;

4. in the continuum limit along the trajectory,

$$m \geq \frac{\hbar c \min(\gamma_*, \gamma')}{a_*} > 0.$$

Proof. (1) is the mixing-to-decay theorem applied to $\Phi^{(K)}$, which lies in the verified neighbourhood by H1 and H2. For (2), write a fine observable as its coarse part plus fluctuation parts step by step, $F = \bar{F}^{(K)} + \sum_{k < K} (F^{(k)} - \bar{F}^{(k)})$; the coarse-part correlations decay by (1) at rate γ' per coarse unit, that is $\gamma'/2^K$ per fine unit, and the fluctuation-part correlations at step k decay by H1 at rate γ_* per unit of $2^k a_0$, that is $\gamma_*/2^k \geq \gamma_*/2^K$ per fine unit; the cross terms are bounded by the same rates. (3) is link reflection positivity together with the spectral argument of the Wilson note §3: decay at rate μ per fine unit for all local observables forces the spectral measure of every local vector orthogonal to the vacuum to vanish above $e^{-\mu}$. (4) follows from (3) as $a_0 \rightarrow 0$ with $2^{K(a_0)} a_0 \rightarrow a_*$. $\square$

The Jaffe–Witten clause $0 < m < \infty$ is then (4) together with the upper side of the moment-hierarchy note; T3, the ratio $m/(\hbar c \Lambda)$, is the statement that $a_* \Lambda$ has a limit, which is the convergence of the trajectory to one curve as in the finite-verification note.

4. What the theorem makes visible

The bound is a ratio of a certified rate to a physical length. With $a_* \simeq 0.093$ fm at $\beta_W = 6$ and the measured glueball mass $1.66 \hbar c \text{ fm}^{-1}$, the true rate is $ma_* \simeq 0.78$; a certified γ' of any size gives a theorem, and one of order 0.1 would already be a physically meaningful bound.

Both hypotheses are finite in kind. H2 is one box at one coupling. H1 is a sequence of steps, uniform in the perturbative regime, whose non-perturbative part is three steps. Neither is available today: H2 needs a certified computation, and H1's last three steps are the confinement-scale problem.

Nothing else is missing. T1, T2, the strong-coupling side below the verified coupling, the transfer from decay to gap, the upper side of the gap, and the reduction of T3 are all in place with explicit constants. The theorem is the repository's formal endpoint: a proof of the mass gap for $SU(3)$ consists of H1 and H2.

5. Consequence for STATE

The map is a theorem with two hypotheses and an explicit conclusion $m \geq \hbar c \min(\gamma_*, \gamma')/a_*$. Progress from here is progress on H1 or H2, and nothing else.